

JKBOSE 2026

CLASS 11TH

CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES

LECTURE-2

CHEMISTRY

VISHAL SIR

Bhaiichara
ON
TOP.



JK Bole Wallah

Topics To Covered

1 BRIEF HISTORY OF PERIODIC TABLE

2 MENDELEEVES PERIODIC TABLE



REVISION.

Stability of half filled
and fully filled orbitals.



a) Symmetry

b) Exchange energy.

$$\frac{n(n-1)}{2}$$

1) Significance of
classification.

BRIEF HISTORY.

a) DOBERINER TRIAD.

Li	Ca	Cl
Na	Sr	Br
K	Ba	I

b) NEWLAND'S LAW OF OCTAVES.

c) Lothar Meyer Arrangement

d) MENDELEEV P. T.

e) Modern PERIODIC TABLE.

BRIEF HISTORY OF CLASSIFICATION OF ELEMENTS



1. Dobereiner's triads

- In 1817, a German scientist, Johann Dobereiner classified the elements in groups of three elements called triads. The elements in a triad had similar properties and the atomic weight of the middle member of each triad is very close to the arithmetic mean (average) of the other two elements. The common triads of Dobereiner classification were:
:lithium, sodium and potassium;
calcium, strontium and barium;
chlorine, bromine and iodine; etc.

Element	Atomic weight	
Lithium Sodium Potassium	7 23 39	Mean of 1st and 3rd $\frac{7 + 39}{2} = 23$
Calcium Strontium Barium	40 88 137	Mean of 1s and 3rd $\frac{40 + 137}{2} = 88.5$
Chlorine Bromine Iodine	35.5 80 127	Mean of 1st and 3rd $\frac{35.5 + 127}{2} = 81.25$

BRIEF HISTORY OF CLASSIFICATION OF ELEMENTS



2. Newlands law of octaves:

- In 1865, an English chemist, John Newlands proposed a new system of grouping elements of similar properties. "According to him, when the elements are arranged in the increasing order of atomic weights, the properties of every eighth element are similar to the first one."
- This relationship is just like every eighth note resembles the first in octaves of music. The Newlands law of octaves seemed to be true only for elements up to calcium as shown below:

Element	Li	Be	B	C	N	O	F
At. wt.	7	9	11	12	14	16	19
Element	Na	Mg	Al	Si	P	S	Cl
At. wt.	23	24	27	29	31	32	35.5
Element	K	Ca					
At. wt.	39	40					

~~H~~ ~~He~~

Li	Be	B	C	N	O	F
Na	Mg	Al	Si	P	S	Cl
K	Ca					

~~Ne~~

~~Ar~~

MUSICAL
OCTAVE

SA

RE

GA

MA

PA

DA

NI

SA.
—

BRIEF HISTORY OF CLASSIFICATION OF ELEMENTS



3. Lothar Meyer's arrangement:

- In 1869, a German chemist, **Lothar Meyer** used the physical properties such as atomic volume, melting point, boiling point, etc., to arrive at his table of elements. He showed that when the properties of the elements such as atomic volume, melting point, boiling point, etc. are plotted as a function of their atomic weights, the elements with similar properties occupied almost similar positions.

Atom
Volume

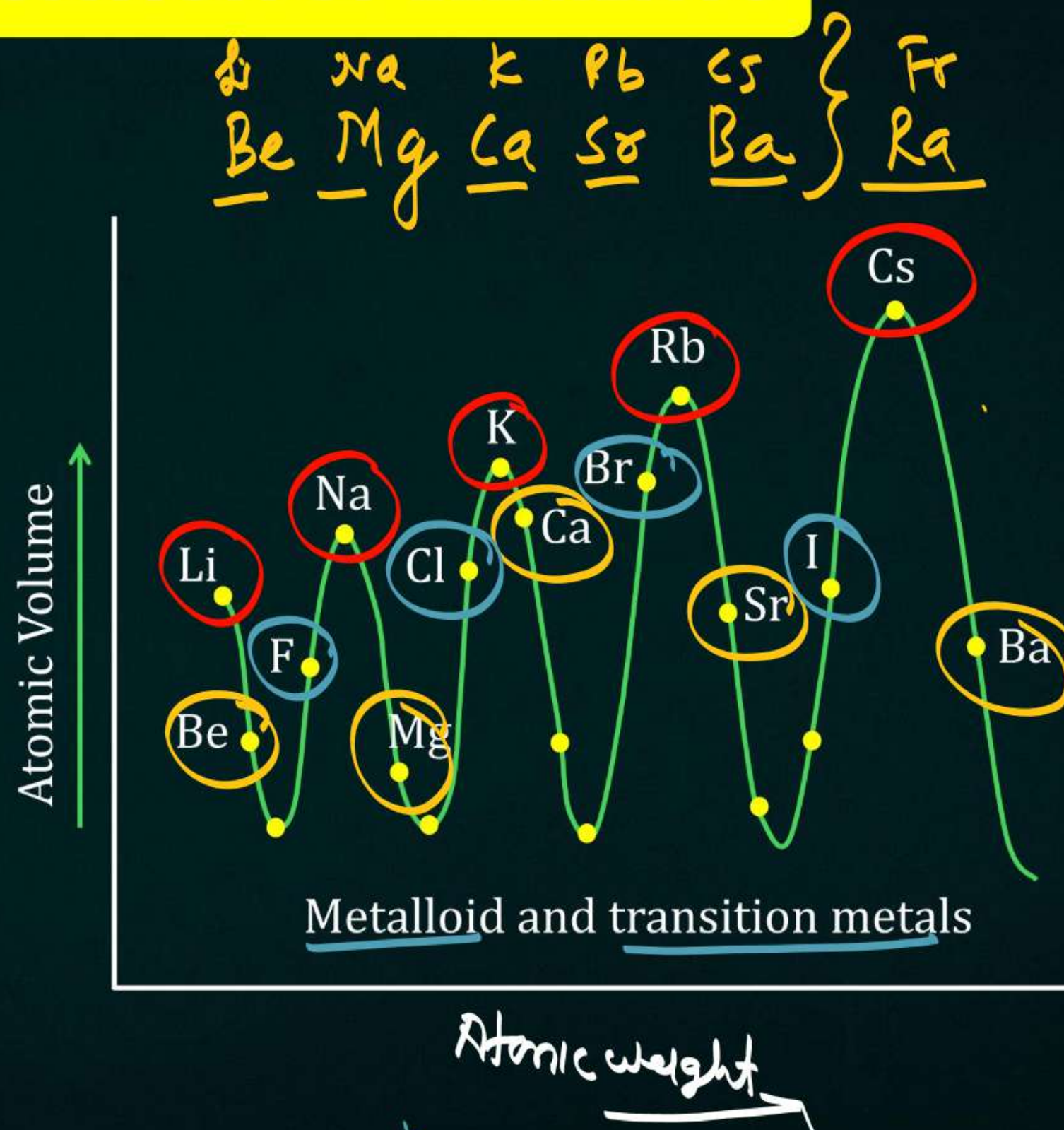
Atomic weight.

BRIEF HISTORY OF CLASSIFICATION OF ELEMENTS



For example,

- (i) The most strongly electropositive alkali metals (Li, Na, K, Rb and Cs) occupy the peaks on the curve.
- (ii) The less strongly electropositive alkaline earth metals (Be, Mg, Ca, Sr and Ba) occupy the descending positions on the curve.
- (iii) The most electronegative elements ie., halogens (F, Cl, Br and I) occupy the ascending positions on the curve.



BRIEF HISTORY OF CLASSIFICATION OF ELEMENTS



JAADU GAR.

4. Mendeleev's Periodic Law and table:

In 1869, Mendeleev made a remarkable contribution to the classification of elements. On the basis of physical and chemical properties of the elements, he gave a law known as the periodic law. (Mendeleev's PERIODIC LAW).

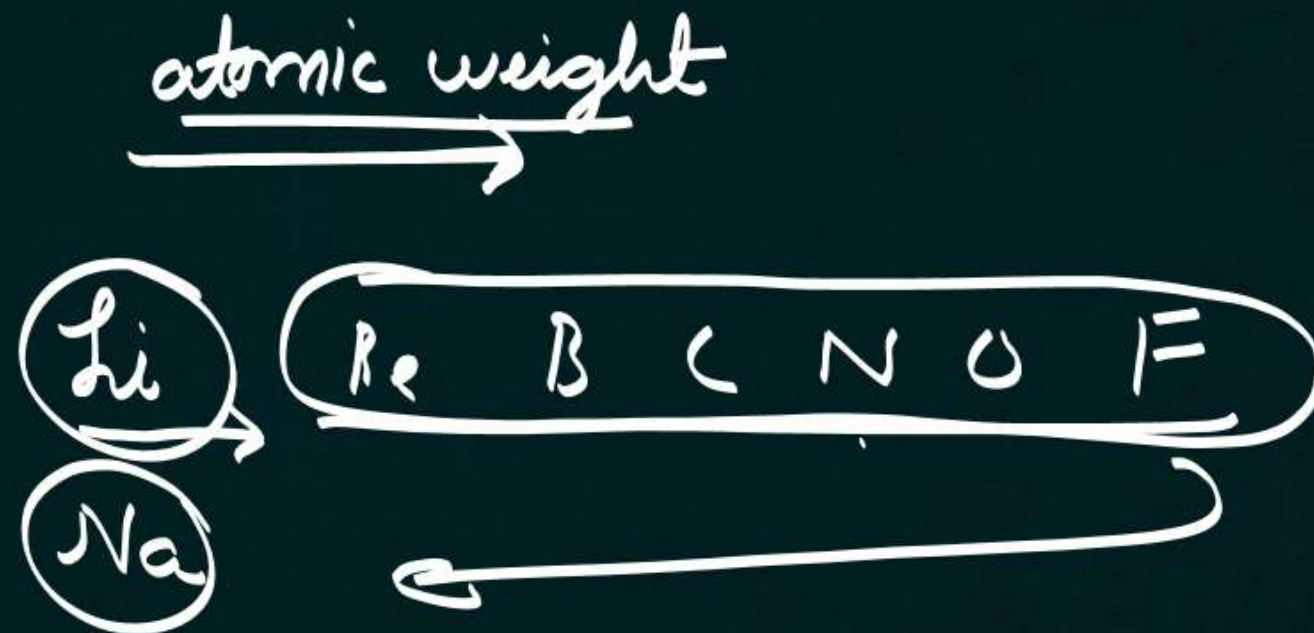
“The law states that the physical and chemical properties of elements are periodic function of their atomic weights.”

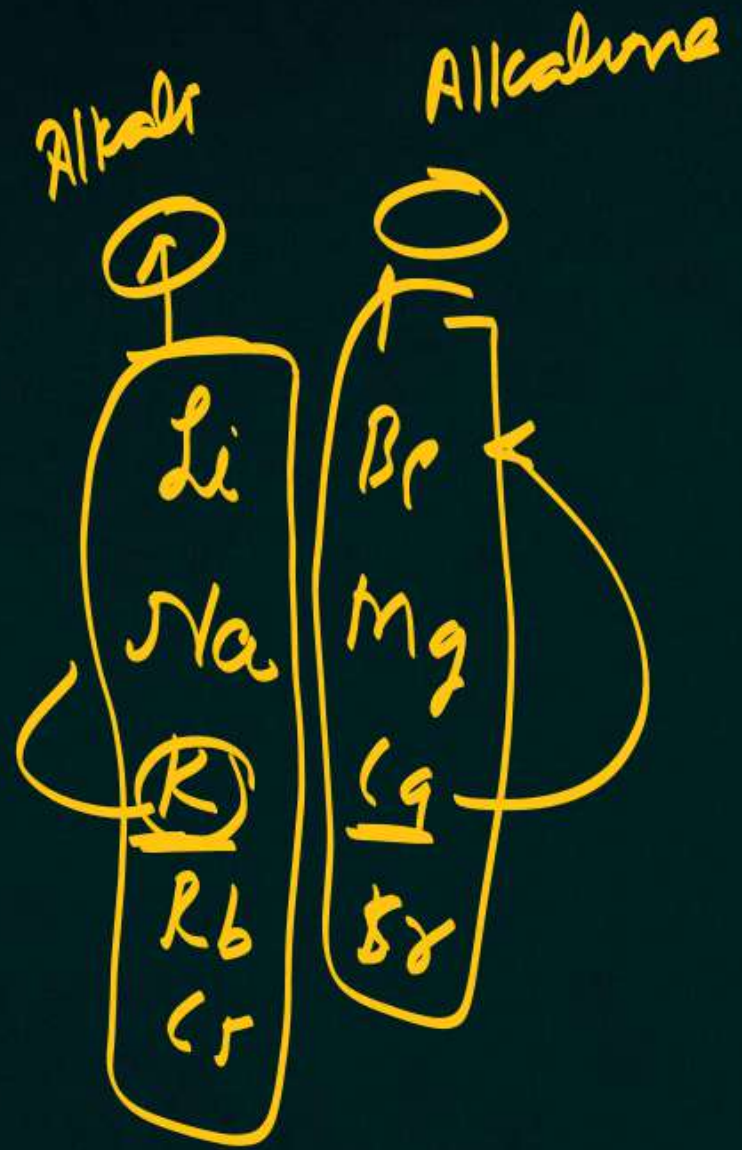
1. Mendeleev arranged elements by increasing atomic weights, grouping those with similar properties.
2. He left gaps for unknown elements, predicting their future discovery.
3. He forecasted the properties of unknown elements based on group similarities.
4. He predicted the properties of eka-aluminium (discovered as gallium) and eka-silicon (discovered as germanium).
5. The discovered elements matched Mendeleev's predictions, proving his table's accuracy (e.g., Winkler's findings on germanium).

↑ Sanskrit word → First
 (Eka) - Aluminium → Ga.
Eka - Silicon → (Ge.)

PERIODIC FUNCTION.

↓
Repetition of elements with
 similar properties after
 regular intervals.





BRIEF HISTORY OF CLASSIFICATION OF ELEMENTS



Characteristics of mendeleevs periodic table:

Mendeleev's periodic table consists of:

- (i) Eight vertical columns called groups. These are designated as I, II, III, IV, V, VI, VII and VIII. Except for group VIII, each group is further divided into two sub-groups designated as A and B. Group zero consists of inert gases.
- (ii) Six horizontal rows called series or periods. These are numbered from 1 to 6.

BRIEF HISTORY OF CLASSIFICATION OF ELEMENTS



Groups	I	II	III	IV	V	VI	VII	VIII			
Oxides Hydrides	RO RH	RO RH ₂	R ₂ O ₃ RH ₃	RO ₂ RH ₄	R ₂ O ₅ RH ₃	RO ₃ RH ₂	R ₂ O ₇ RH	RO ₄			
Periods ↓	A	B	A	B	A	B	A	B	Transition series		
1	H 1.808										
2	Li 6.939	Be 6.939	B 6.939	C 12.011	N 14.007	O 15.999	F 18.998				
3	Na 22.99	Mg 24.31	Al 29.98	Si 28.09	P 30.974	S 32.06	Cl 35.453				
4 First series Second series	K 39.102 Cu 63.54	Ca 40.08 Zn 65.37	Sc 44.96 Ga 69.72	Ti 47.90 Ge 72.59	V 50.94 As 74.92	Cr 50.20 Se 78.96	Mn 54.94 Br 79.909	Fe 55.85	Co 58.93	Ni 58.71	
5 First series Second series	Rb 85.47 Ag 107.87	Sr 87.62 Cd 112.40	Y 88.91 In 114.82	Zr 91.22 Sn 118.69	Nb 92.91 Sb 121.75	Mo 95.94 Te 127.60	Tc 99 I 126.90	Ru 101.07	Rh 102.91	Pd 106.4	
6 First series Second series	Cs 132.90 Au 196.97	Ba 137.34 Hg 200.59	La 138.91 Tl 204.37	Hf 178.49 Pb 207.19	Ta 180.95 Bi 208.98	W 183.85		Os 190.2	Ir 192.2	Pt 135.09	

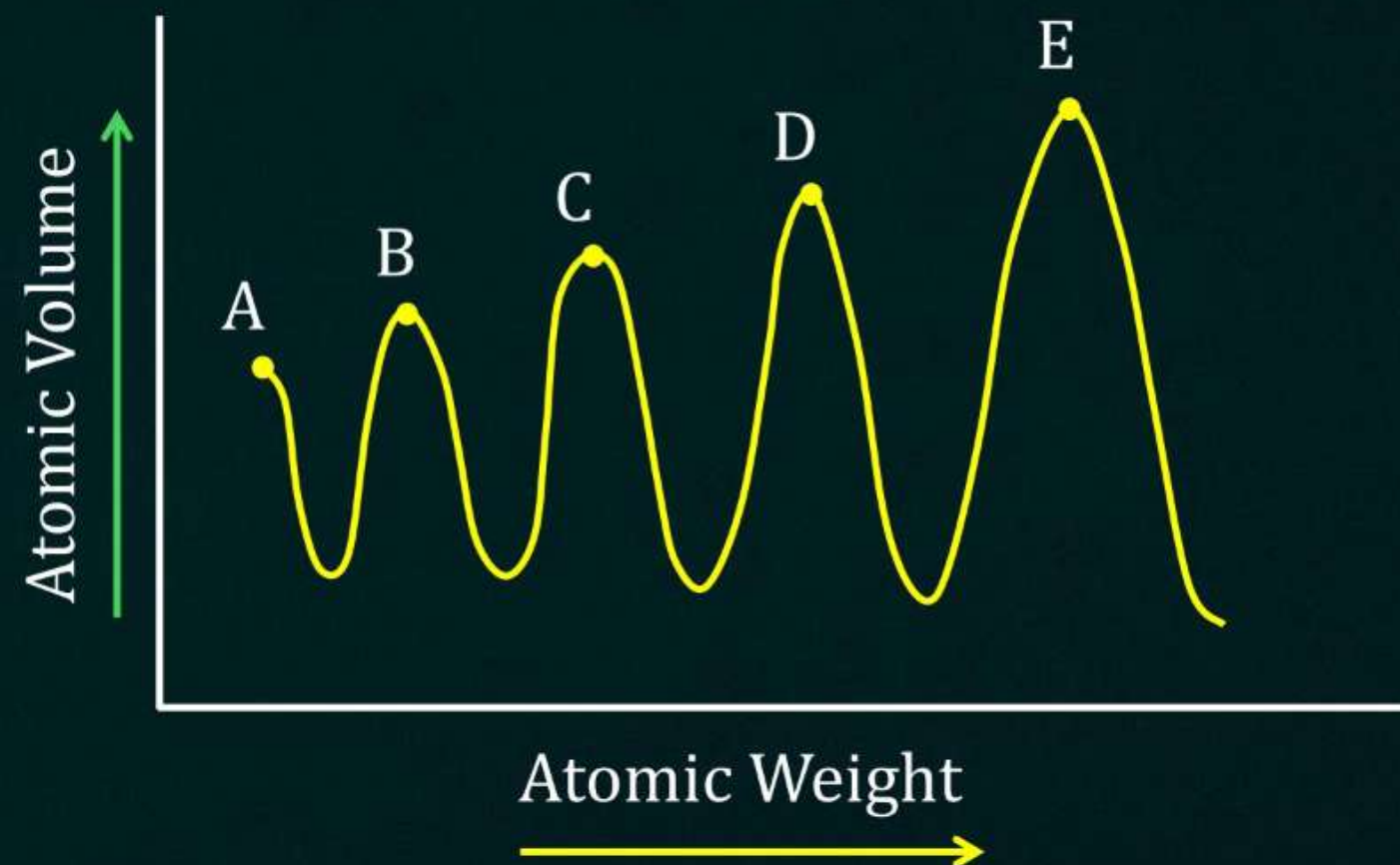
$\xrightarrow{15} 16 \quad 18(\underline{\quad}) \quad (24)$

QUESTION



In the Lothar – Mayer graph A, B, C, D, & E element are:

- 1 Halogens
- 2 Alkaline earth metals
- 3 Alkali metals
- 4 Transition metals



QUESTION



Which important property did Mendeleev use to classify the element in his periodic table?

- 1** Atomic weight
- 2** Atomic number
- 3** Melting point
- 4** None of these

QUESTION



Which of the following statements is incorrect? (*False*)



- 1 Mendeleev's arranged elements in horizontal rows and vertical columns 
- 2 Mendeleev's arranged elements in order of their increasing atomic number *False*
- 3 Mendeleev's system of classifying elements was more elaborate than that of Lothar Meyer 
- 4 None of the above

QUESTION



Mendeleev's left the gap under aluminium and a gap under silicon having atomic weights 68 and 72 respectively. These elements respectively are

Eka - Al $\rightarrow$ 
Eka $\rightarrow$ Si $\rightarrow$ 

1 Gallium and Boron

2 Gallium and Germanium

3 Germanium and Silicon

4 Boron and Silicon

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REVISION KO ROZ KARNA NA.

Thank
You